Telecom Network Intelligence System - Design Document

Problem Statement

This project focuses on predicting network congestion risk in telecom regions based on usage patterns such as calls, SMS, and internet traffic. Congestion occurs when network demand exceeds capacity, leading to degraded service quality. Identifying congestion risk allows telecom operators to optimize network resources and improve user experience. Since the dataset does not contain labels, a rule-based approach is used where usage above the 90th percentile is labeled as HIGH risk, between 60th and 90th percentile as MEDIUM, and below 60th percentile as LOW.

Design Challenges and Solutions

Major challenges included handling large-scale data, ensuring data quality, efficient database loading, and generating sufficient ML training data. These were addressed by limiting dataset size to 200,000 rows for performance, cleaning data during processing, batching database inserts, and engineering features using region and time combinations.

Batch vs Streaming Classification

Network logs: Streaming. Daily summaries: Batch. Monthly reports: Batch. Congestion alerts: Streaming. Dashboard data: Hybrid batch and near real-time.

Storage Design Justification

Raw data is stored in CSV format due to its simplicity and interoperability. Processed data is stored in Parquet format because it provides columnar storage, efficient compression, and faster analytics queries. Data is partitioned by date to improve query performance. The warehouse layer uses a star schema to separate fact and dimension tables, which improves query efficiency and reduces redundancy.

Failure Handling

The pipeline is designed to handle failures such as missing input files, corrupt rows, and duplicate files. Missing files are logged and skipped without crashing the system. Corrupt rows are filtered during data processing. Duplicate files are managed through validation checks to avoid duplication in the warehouse.

End-to-End Flow

The system processes data from ingestion to visualization. Raw data is processed using Spark, stored in Parquet format, loaded into a MySQL warehouse, accessed through FastAPI, enhanced with ML predictions, and visualized using a React dashboard.